

Dhairya Jain

📧 jaindhairya@iitb.ac.in

📞 +91-7987697281

✉️ jaindhairya2001@gmail.com

🌐 [linkedin.com/in/jain-dhairya/](https://www.linkedin.com/in/jain-dhairya/)

Education

- **Indian Institute of Technology, Bombay** — JEE-Advanced : **1886**, JEE-Mains : **1209** (2018 - 2022)
- *Bachelors* in **Economics (Dean's List)** with a *Minor Degree* in **Operations Research** — CGPA: 8.8/10.0

Work Experience

Microsoft | SWE II (L-62) | Distributed Systems and Infrastructure Engineer (June 2022 - Present)

Azure Storage	- Enhancing the latest scaled-account feature & SKU conversion enabling 6000+ PB-sized accounts - Contributing to high-throughput feature and garbage collection processes for Block-Blob & Append-Blob - Reduced Get-Blob API latency from 240μs to 40μs (83% reduction) to support AI and HPC workloads, by implementing C++ object pooling and lock-free queues to eliminate heap contention and request-path hops - Led the (un-)virtualization feature for scaled GRS accounts, ensuring data integrity using TLA+ specs, and orchestrating components involved in request hot-path across all of data, control & customer planes - Engineered a Secondary Indexing pipeline to transform primary-schema file tables into optimized secondary structures; implemented a Transformer-Collator architecture to process metadata with 100% idempotency - Authored a deterministic state-syncing protocol with a probabilistic convergence guarantee, validated via TLA+; Reducing P99 migration time by 2.7x & eliminating race conditions in distributed state machines Directing technical execution for a 3-person sub-team, including mentoring a junior engineers, conducting architectural audits and deep-dive code reviews to ensure system-wide data integrity and service resiliency
Bing Platform	- Engineered ETL pipelines based on <i>Apache Spark</i>, to develop hardware forecasting solutions over big-data - Designed experiments to find the <i>Cost To Serve</i> per Bing Query, and instrument Bing.exe for un-used code
Tech Stack	▪ C++, C#, Virtualization, Scripting (Python, Powershell, KQL), Databases, TLA+, GDB, Systems / Core Infra

WorldQuant Research Pvt. Ltd. | Quantitative Researcher & Developer Intern (May 2021 - July 2021)

Worked on 2 projects in the areas of **quantitative development** using Cython, and **quantitative research** using machine learning

Quant Dev	- Designed & implemented a high-efficiency command-line tool & Python package to visualize & analyze alphas - Achieved 100x speed-up by integrating a C++ backend with a Cython-based Python interface
Quant Research	- Researched latest financial & machine learning literature to identify fresh ideas for Portfolio Optimization - Designed Python APIs to create, back-test & compare different ideas of aggregating multiple alpha signals
Tech Stack	▪ C++, Cython, Concurrency, Multi-threading, Linux, Econometrics, Machine Learning, Investment Research

Key Projects & Leadership Roles

Google Summer of Code'21 | The R project for Statistical Computing (May 2021 - Aug' 2021)

Contributed to **PerformanceAnalytics**, an *R* package used for portfolio risk analysis

Summary	Created functionality for calculating outlier-insensitive risk estimates in the Capital Asset Pricing Model, aiding 120K+ professionals & researchers in making informed investment choices. Published as — CRAN vignette
Tech Stack	▪ R, Quantitative Finance, Portfolio Management, Multi Factor Models, Robust Regression (Huber, MM etc.)

Deep learning in Symbolic Mathematics | Research Project (Mar' 2022 - Apr' 2022)

- Extended the research done by Facebook AI to learn and predict **taylor series** transformations of mathematical expressions
- Generated symbolic function dataset using **randomized binary trees**; optimized the algorithm using **dynamic programming**
- Experimented with **bi-directional LSTM**-based **encoder-decoder** network and **attention**-based **transformer** models

Web Team Manager | Student Alumni Relations Cell | IIT-Bombay (May 2019 - Apr' 2021)

Summary	- Led a 5-membered team in a 60+ membered body working closely with the Dean & Alumni Association - Administered & maintained 10+ domains of the body using Nginx reverse proxy over Gunicorn web-servers - Eliminated server infrastructure costs by 54% by optimizing application storage and compute resources
Impact	▪ Benefited 3k+ students by ideating 9+ events over the year having a combined budget of INR 1.5+ Million

Achievements

- Endurance Athlete** : Finished multiple triathlons like Bergman Mysuru Olympic 2026, Ironman 5150 Chennai 2026